Problems: Impulse Response Properties and Convolution

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Impulse Response and LTI System Properties

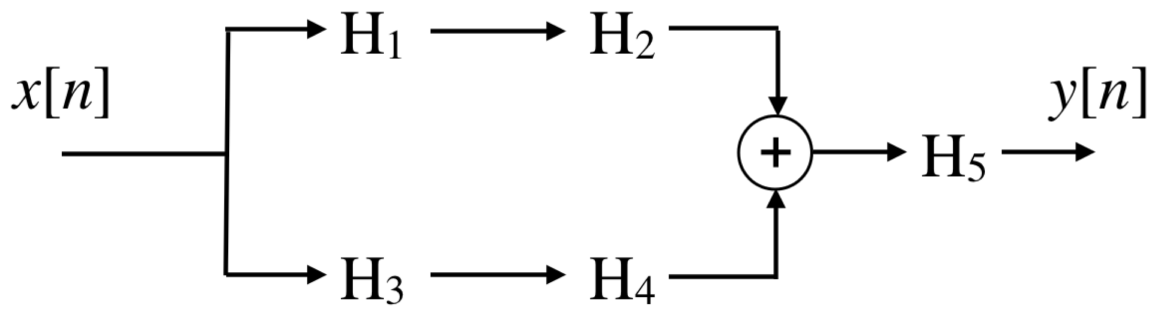
1. A discrete-time system has impulse response $h[n] = (-1)^n u[n-3]$.

   a) Is this system memoryless?

   b) Is this system BIBO stable?

   c) Is this system causal?

2. In the system shown below H_i is an LTI system with impulse response $h_i[n[$.



Which of the following represents the impulse response $h[n]$ for the equivalent system relating $y[n]$ to $x[n]$? Select all that apply.

i) can't be determined from information given

ii) $h[n] = (h_1[n] * h_2[n] + h_3[n] * h_4[n]) * h_5[n]$

iii) $h[n] = h_1[n] * h_2[n] * h_5[n] + h_3[n] * h_4[n] * h_5[n]$

iv) $h[n] = h_5[n] * (h_3[n] * h_4[n] + h_1[n] * h_2[n])$

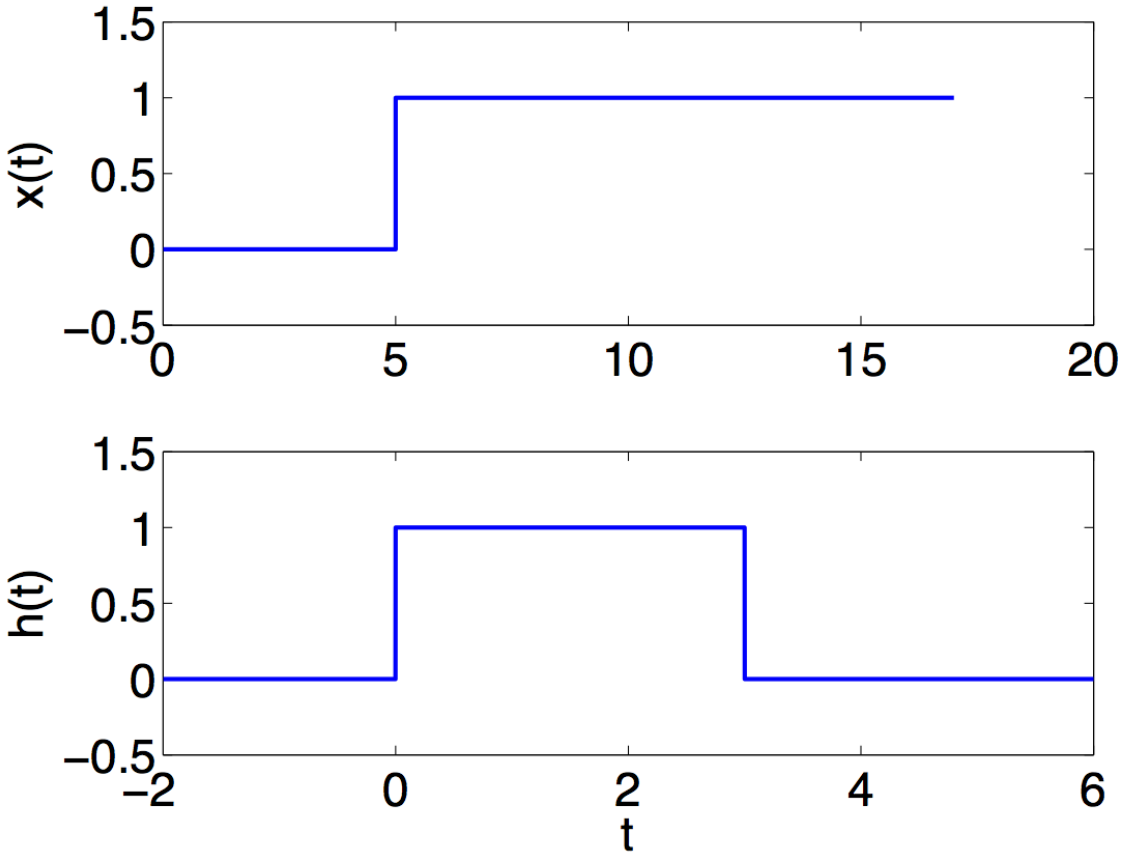
v) $h[n] = h_1[n] * h_5[n] * h_2[n] + h_5[n] * h_4[n] * h_3[n]$

vi) $h[n] = (h_1[n] * h_5[n] + h_5[n] * h_4[n]) * (h_2[n] + h_3[n])$

vii) $h[n] = (h_1[n] + h_2[n]) * (h_3[n] + h_4[n]) + h_5[n]$

Convolution

3. In this problem we consider the convolution of continuous-time signals $x(t)$ and $h(t)$ where $x(t) = u(t-5)$ and $h(t) = u(t) - u(t-3)$ as shown below.



Complete the following statements:

a) Consider $x(-\tau)$. $x(-\tau) = 0$ for τ _____

b) Consider $x(t-\tau)$. If $t=3$, then $x(3-\tau)=0$ for τ _____

c) Consider $x(t-\tau)$. If $t=5$, then $x(5-\tau)=0$ for τ _____

d) Consider $x(t-\tau)$. If $t=7$, then $x(7-\tau)=0$ for τ _____

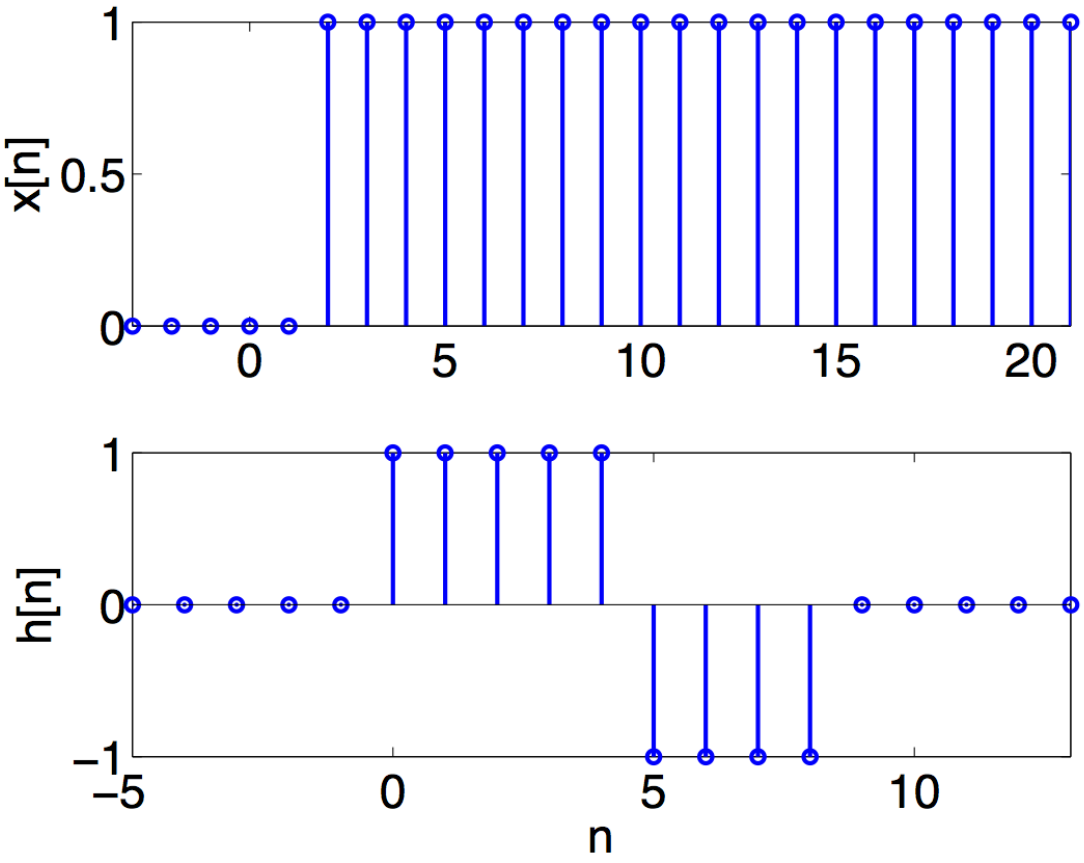
e) Now consider $h(\tau)x(t-\tau)$. For $h(\tau)x(t-\tau)\neq 0$ for some τ, we must have t _____

f) Now consider $y(t)=\int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$ and complete the statement $y(t)=0$ for t _____

g) As t increases, $y(t)$ is non zero and constant if t is _____

h) As t increases, $y(t)$ (increases, decreases) provided t is greater than _____ and less than _____

4. In this problem we consider the convolution of discrete-time signals $h[n] = u[n] - 2u[n-5] + u[n-9]$ and $x[n] = u[n-2]$ as shown below.



Complete the following statements:

a) Consider $x[-k]$. $x[-k] = 0$ for k _____

b) Consider $x[n-k]$. If $n=-1$, then $x[-1-k]=0$ for k _____

c) Consider $x[n-k]$. If $n=1$, then $x[1-k]=0$ for k _____

d) Consider $x[n-k]$. If $n=3$, then $x[3-k]=0$ for k _____

e) Consider $x[n-k]$. If $n=7$, then $x[7-k]=0$ for k _____

f) Now consider $x[n-k]h[k]$. The condition $x[n-k]h[k] \neq 0$ for some k, requires n be _____

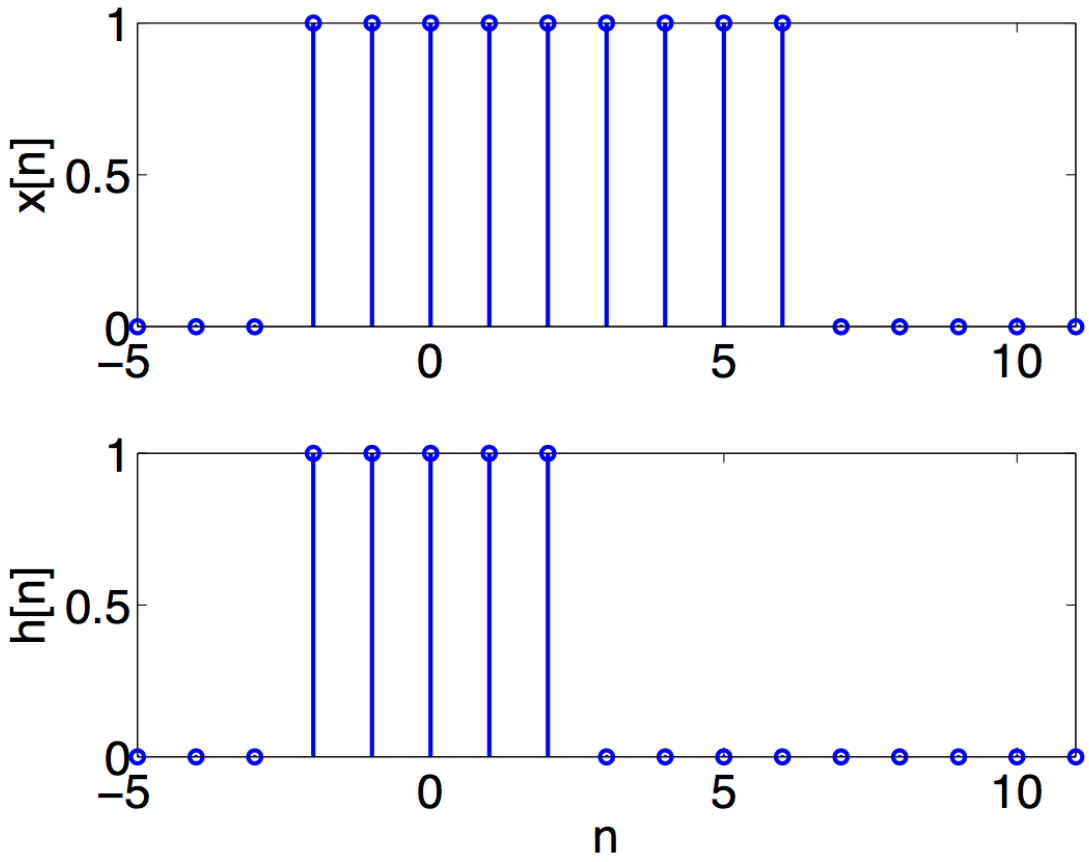
g) Now consider $y[n]=\sum_{k=-\infty}^{\infty} x[n-k]h[k]$ and complete the statement $y[n]=0$ for n _____

h) As n increases, $y[n]$ remains constant and non zero provided n is _____

i) Consider the interval $2 \leq n \leq 6$. As n increases, $y[n]$ (increases, decreases, is constant).

j) Consider the interval $6 \leq n \leq 9$. As n increases, $y[n]$ (increases, decreases, is constant).

5. Consider the convolution of discrete-time signals $x[n] = u[n+2] - u[n-7]$ and $h[n] = u[n+2] - u[n-3]$ as shown below.



Complete the following statements. Hint - consider the product $x[n-k]h[k]$ for different values of n, and remember convolution is the sum of this product over all values of k: $y[n] = \sum_{k=-\infty}^{\infty} x[n-k]h[k]$.

a) For n less than _____ $y[n] = 0$.

b) As n increases beyond this point, $y[n]$ (increases, decreases, is constant) until $n =$ _____

c) As n continues to increase, $y[n]$ (increases, decreases, is constant) until $n =$ _____

d) As n continues to increase, $y[n]$ (increases, decreases, is constant) until $n =$ _____

e) As n continues to increase beyond this point, $y[n]$ (increases, decreases, is constant).

f) For $n > 100$, the value of $y[n]$ is _____

6. Suppose an LTI system has impulse response $h(t) = e^{-t^2}\cos(\pi t)u(t+3)$. What is the output of this system $y(t)$ if the input $x(t) = 2\delta(t-2)$? Hint: use linearity, time invariance, and the meaning of the impulse response.

a) $y(t) = 2e^{-t^2}\cos(\pi t)u(t+1)$

b) $y(t) = 2e^{-(t-2)^2}\cos(\pi(t-2))u(t+1)$

c) $y(t) = 2e^{-(t-2)^2}\cos(\pi(t-2))u(t+3)$

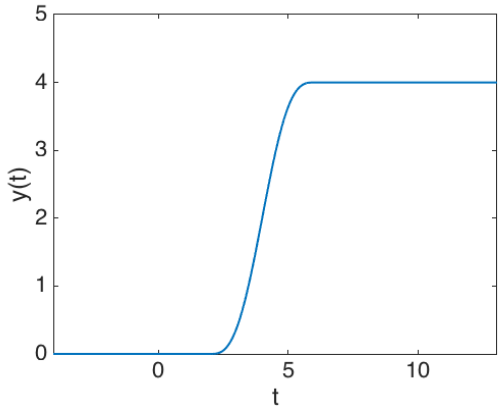
d) $y(t) = 2e^{-(t+2)^2}\cos(\pi(t+2))u(t+5)$

e) $y(t) = 2e^{-(t+2)^2}\cos(\pi(t+2))u(t+3)$

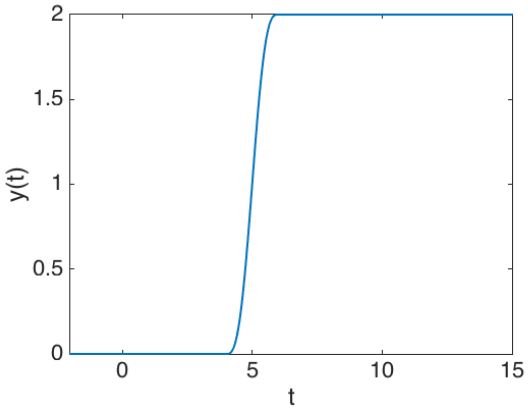
f) $y(t) = 2e^{-t^2}\cos(\pi t)u(t+5)$

g) $y(t) = 2e^{-t^2}\cos(\pi t)u(t+3)$

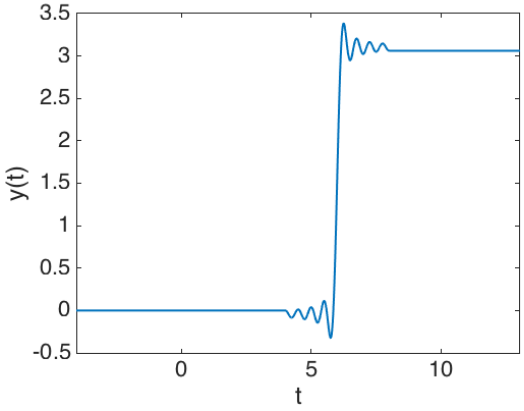
Problems 7-9 ask you to find the output of a continuous-time LTI system given the input and impulse response. The graphs below represent the choices for the output $y(t)$ in each case.



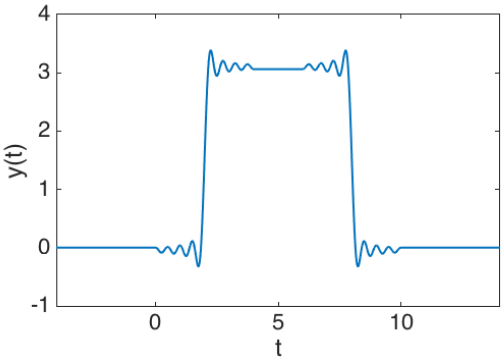
(a)



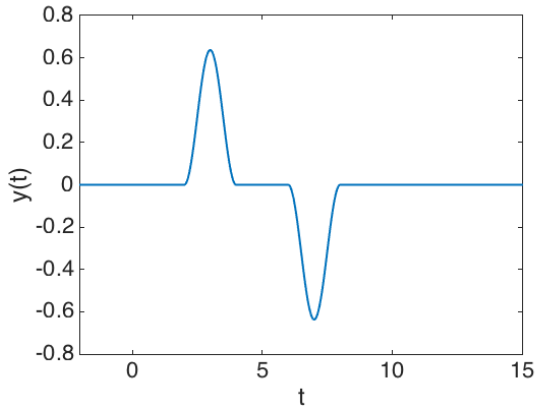
(b)



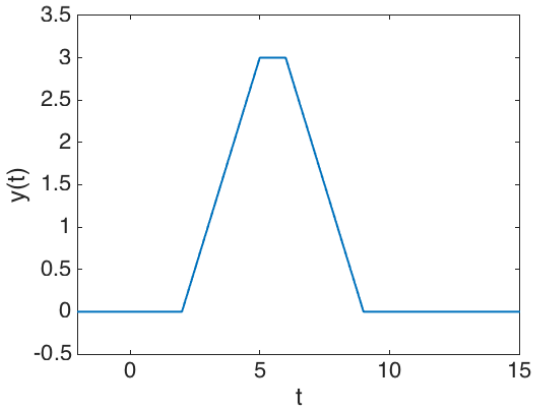
(c)



(d)

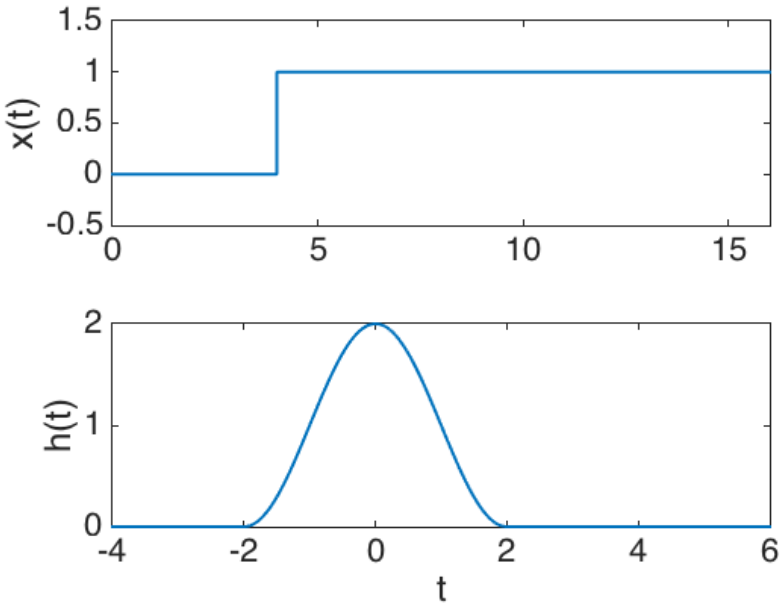


(e)

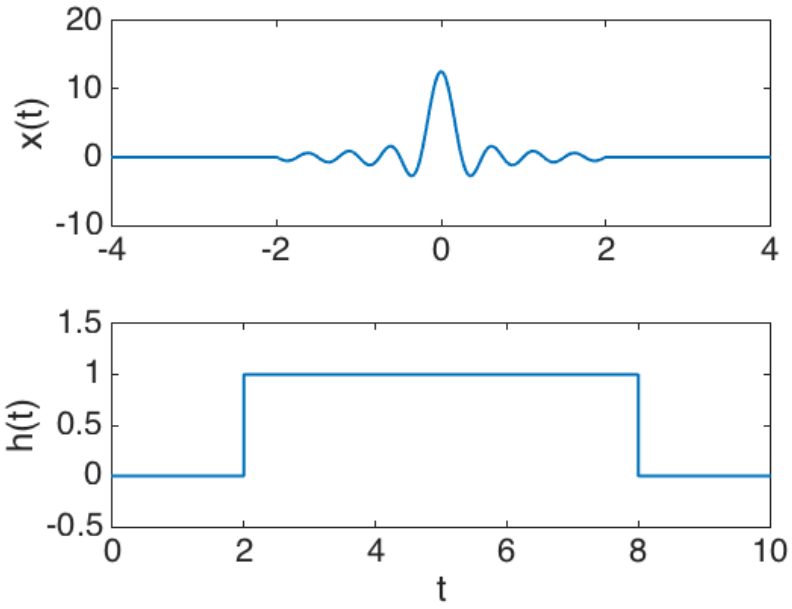


(f)

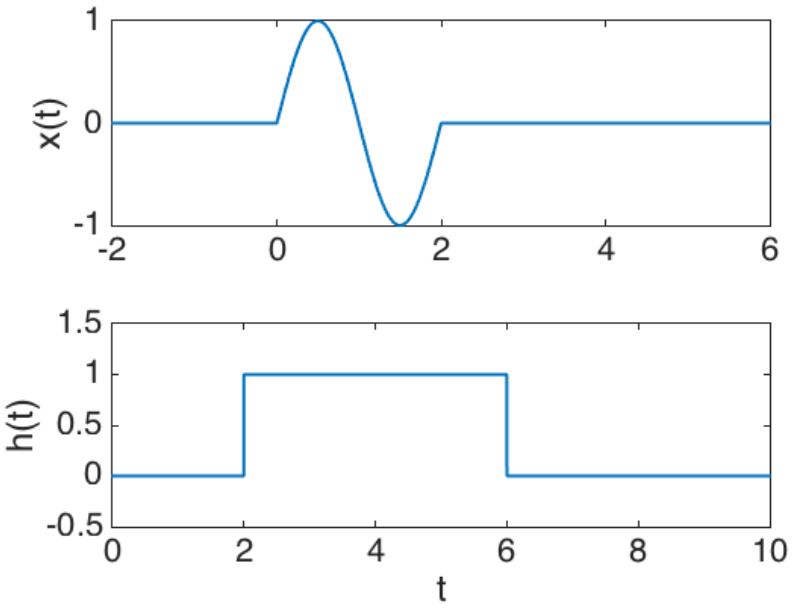
7. Identify the correct output for the input $x(t)$ and LTI system impulse response $h(t)$ depicted below.



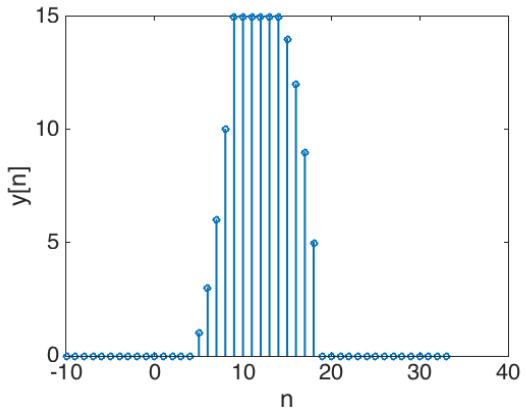
8. Identify the correct output for the input $x(t)$ and LTI system impulse response $h(t)$ depicted below.



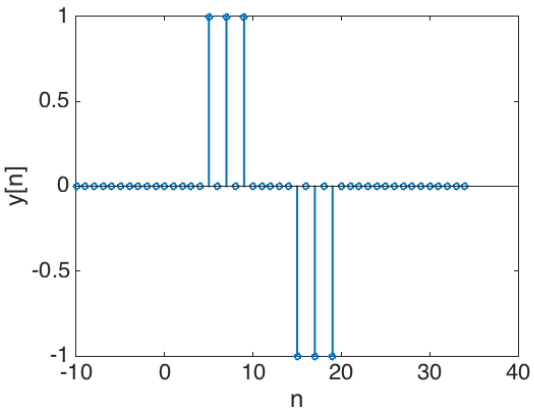
9. Identify the correct output for the input $x(t)$ and LTI system impulse response $h(t)$ depicted below.



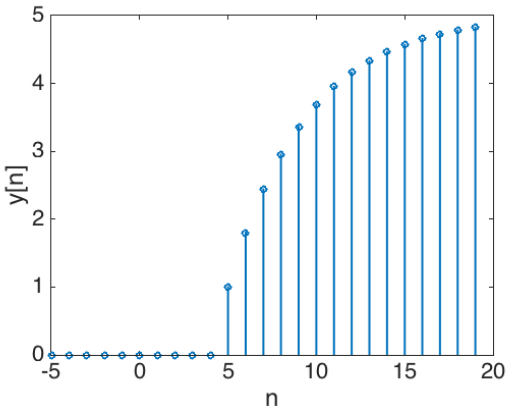
Problems 10-12 ask you to find the output of a discrete-time LTI system given the input and impulse response. The graphs below represent the choices for the output $y[n]$ in each case.



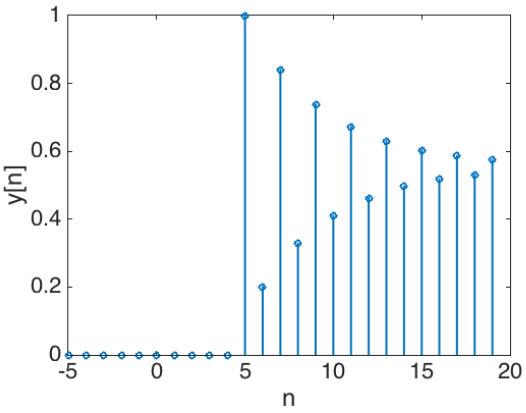
(g)



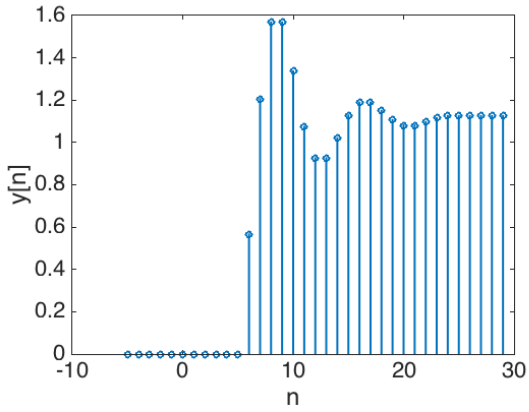
(h)



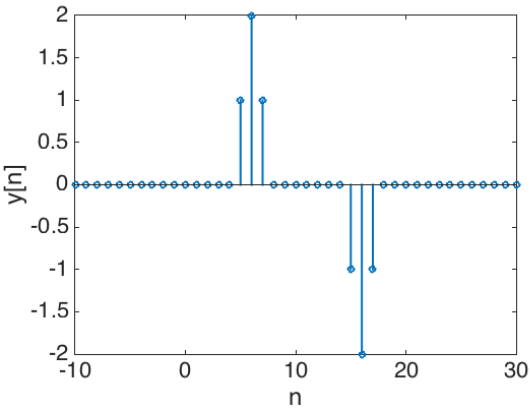
(i)



(j)

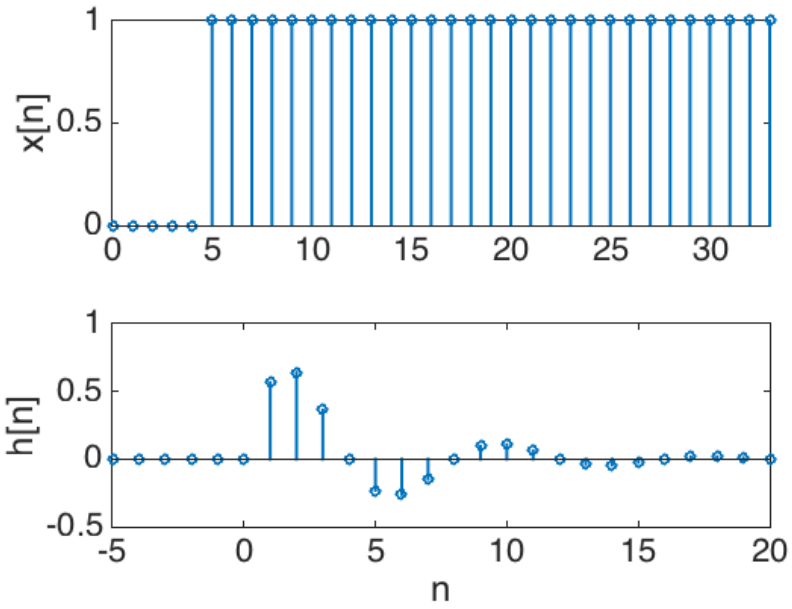


(k)

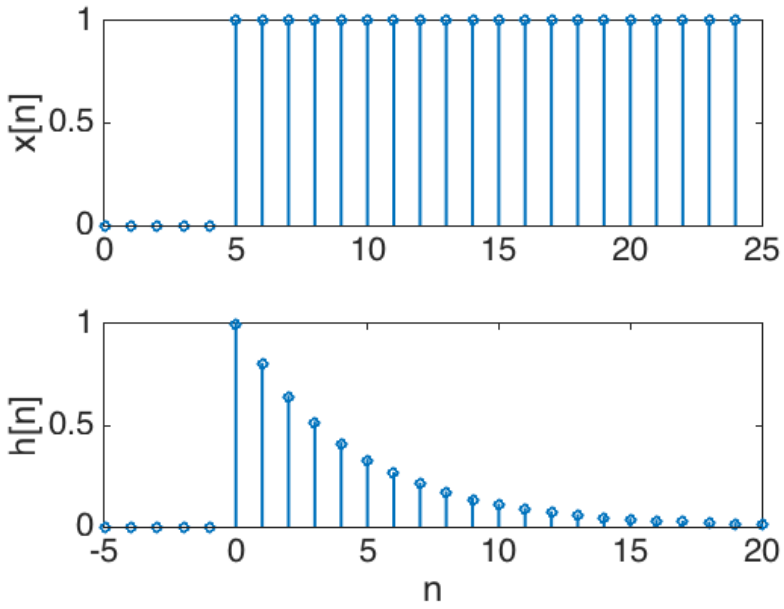


(l)

10. Identify the correct output for the input $x[n]$ and LTI system impulse response $h[n]$ depicted below.



11. Identify the correct output for the input $x[n]$ and LTI system impulse response $h[n]$ depicted below.



12. Identify the correct output for the input $x[n]$ and LTI system impulse response $h[n]$ depicted below.

